

Lab 4 - Variables in Prolog

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Family Facts

```
male(ali).
male(ahmad).
male(muhammad).
male(faisal).
male(hassan).
male(khan).

female(sara).
female(nora).

parent(ali, ahmad).
parent(ali, muhammad).
parent(ahmad, sara).
parent(ahmad, nora).
parent(ahmad, khan).
parent(muhammad, faisal).
parent(faisal, hassan).
```

Task 1 - Unification Queries

For each query, the Prolog result and a brief explanation are provided.

1. Query

```
?- house = house.
```

Result:

```
true.
```

Explanation: Both terms are identical, so they unify successfully.

2. Query

```
?- house = abc.
```

Result:

```
false.
```

Explanation: house and abc are different atoms, so they cannot unify.

3. Query

```
?- edge(a,b) = edge(C,b).
```

Result:

$C = a.$

Explanation: Both structures have the same functor and arity. C is instantiated to a.

4. Query

?- $\text{edge}(a,d) = \text{edge}(C,B).$

Result:

$C = a,$

$B = d.$

Explanation: The structures match, so C is instantiated to a and B to d.

5. Query

?- $\text{edge}(C,B) = \text{edge}(a,d).$

Result:

$C = a,$

$B = d.$

Explanation: The variables are instantiated so that both terms become identical.

6. Query

?- $\text{edge}(A,f(g,h)) = \text{edge}(a,B).$

Result:

$A = a,$

$B = f(g, h).$

Explanation: A unifies with a, and B unifies with the complete structure f(g,h).

7. Query

?- $\text{edge}(A,A) = \text{edge}(b,c).$

Result:

false.

Explanation: The same variable A must have the same value in both positions. It cannot be both b and c.

8. Query

?- $\text{edge}(A,A) = \text{edge}(b,b).$

Result:

$A = b.$

Explanation: Both positions contain b, so the same variable A can unify with both.

9. Query

?- $A = B, B = f.$

Result:

A = B,
B = f.

Explanation: A and B first become linked. Then B is instantiated to f, so A represents the same value.

10. Query

?- A = B, B = f, A = e.

Result:

false.

Explanation: After A and B are linked and B becomes f, A cannot also unify with the different atom e.

Task 2 - Family Queries

Queries are based on the family facts given in the lab.

1. Names of all females

?- female(X).

Result:

X = sara ;
X = nora.

Explanation: The variable X matches each fact declared with female/1.

2. Whether there is any male

?- male(_).

Result:

true.

Explanation: The anonymous variable (_) is used because only the existence of a male is required.

3. Grandparent of Faisal

?- parent(X, Y), parent(Y, faisal).

Result:

X = ali,
Y = muhammad.

Explanation: Muhammad is the parent of Faisal, and Ali is the parent of Muhammad. Therefore, Ali is Faisal's grandparent.

4. All daughters of Ahmad

?- parent(ahmad, X), female(X).

Result:

X = sara ;
X = nora.

Explanation: X must be both a child of Ahmad and female, giving Sara and Nora.

5. Whether Ahmad and Muhammad have a common parent

```
?- parent(X, ahmad), parent(X, muhammad).
```

Result:

```
X = ali.
```

Explanation: Ali is a parent of both Ahmad and Muhammad, so they have a common parent.

6. Whether Khan is the brother of Nora

```
?- male(khan), parent(X, khan), parent(X, nora).
```

Result:

```
X = ahmad.
```

Explanation: Khan is male and Ahmad is a parent of both Khan and Nora, so Khan is Nora's brother.